Prompt Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Prompt Experiments before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Predictive vs Generative

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Text as Pieces

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Nearest Neighbor Reasoning

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Prompt Experiments trace and record the first step you can explain without help.

Use W2: Trace the worked case for Controlled **prompt** experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Prompt Experiments explanation against the supplied facts before discussing it.

Use the worked case for Controlled **prompt** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Prompt Experiments target to a trace and a justified conclusion.

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when **output**s vary across runs.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Prompt Experiments case and show the deciding evidence.

Prompt B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

Design a fair comparison.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Prompt A and B are tested on different questions. B scores higher. Can wording alone explain the difference? Design a **controlled comparison**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 28 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Prompt Experiments adds to Transparent Book Recommender and keep the evidence.

Test generated book explanations with a fixed **rubric** and prohibit invented details.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when **output**s vary across runs.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Prompt Experiments facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Prompt Experiments facts and keep your reasoning.

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Prompt Experiments facts and keep your reasoning.

A report omits this qualification: **Prompt** selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Prompt Experiments facts and keep your reasoning.

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Prompt Experiments facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Prompt Experiments

> A fictional book-help project tests two prompt versions on seven shared development cases. Explain the labelled step before checking the changed condition.

A fictional book-help project tests two **prompt** versions on seven shared development cases.

P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

AI ACADEMY / AI-2/28

Paired overall and missing-source passes

	Overall passes /7	Missing-source passes /2
P	4	1
Q	5	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/28

Prompt Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired setup

Use the same seven cases and pass criteria for both prompts.



2. Overall rates

P=4/7; Q=5/7.



3. Missing-source slice

P passes 1/2; Q passes 0/2 in cases 4 and 7.



4. Decide explicitly

Q improves overall count but worsens missing-source behavior; use the declared priorities.

CHANGE ONE THING

Make missing-source correctness a required selection gate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Name the case P passes and Q fails, and explain why it matters.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-help project tests two **prompt** versions on seven shared development cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Q improves overall count but worsens missing-source behavior; use the declared priorities.

1. Decide explicitly

2. Paired setup

3. Missing-source slice

4. Overall rates

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Decide explicitly Q improves overall count but worsens missing-source behavior; use the declared priorities.

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Make missing-source correctness a required selection gate. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Neither **prompt** clears a 2/2 gate on these records.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Neither **prompt** clears a 2/2

gate on these records. P fails 7 and Q fails both 4 and 7; both need revision.

Repair a missing building block

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

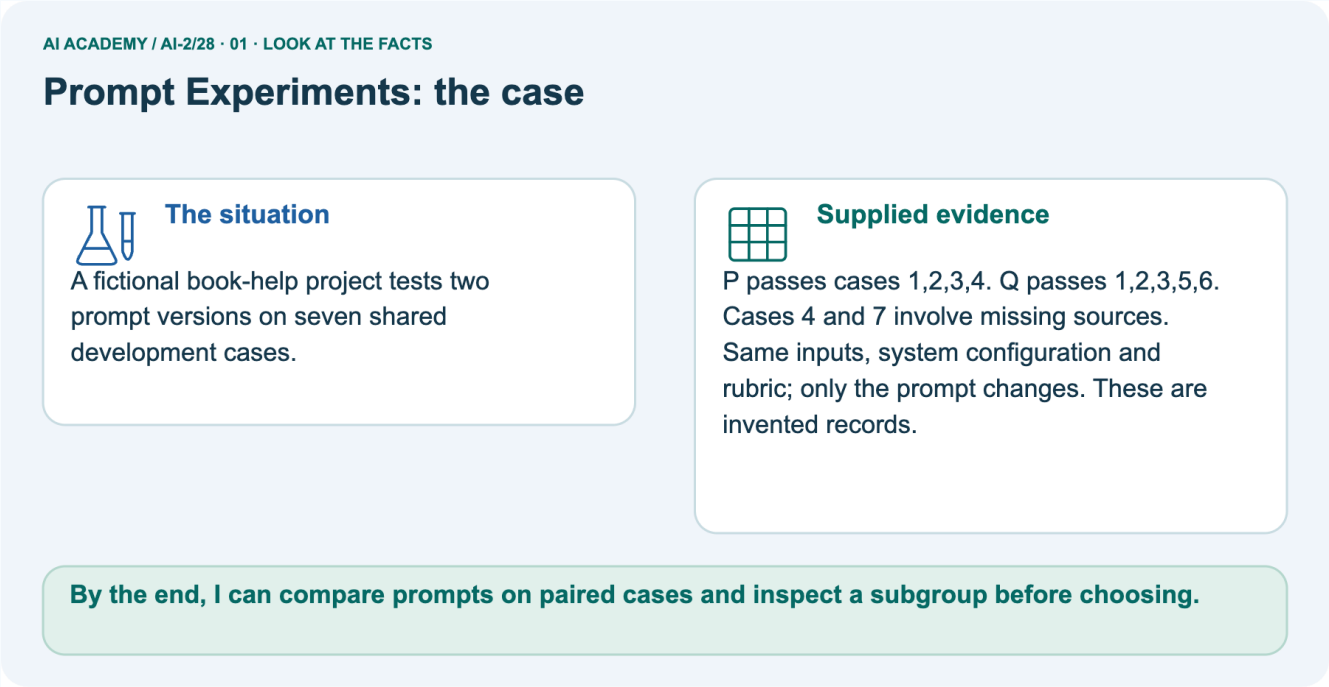
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

Explain each Prompt Experiments picture from its labels, then record a relevant limit.

01. Prompt Experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

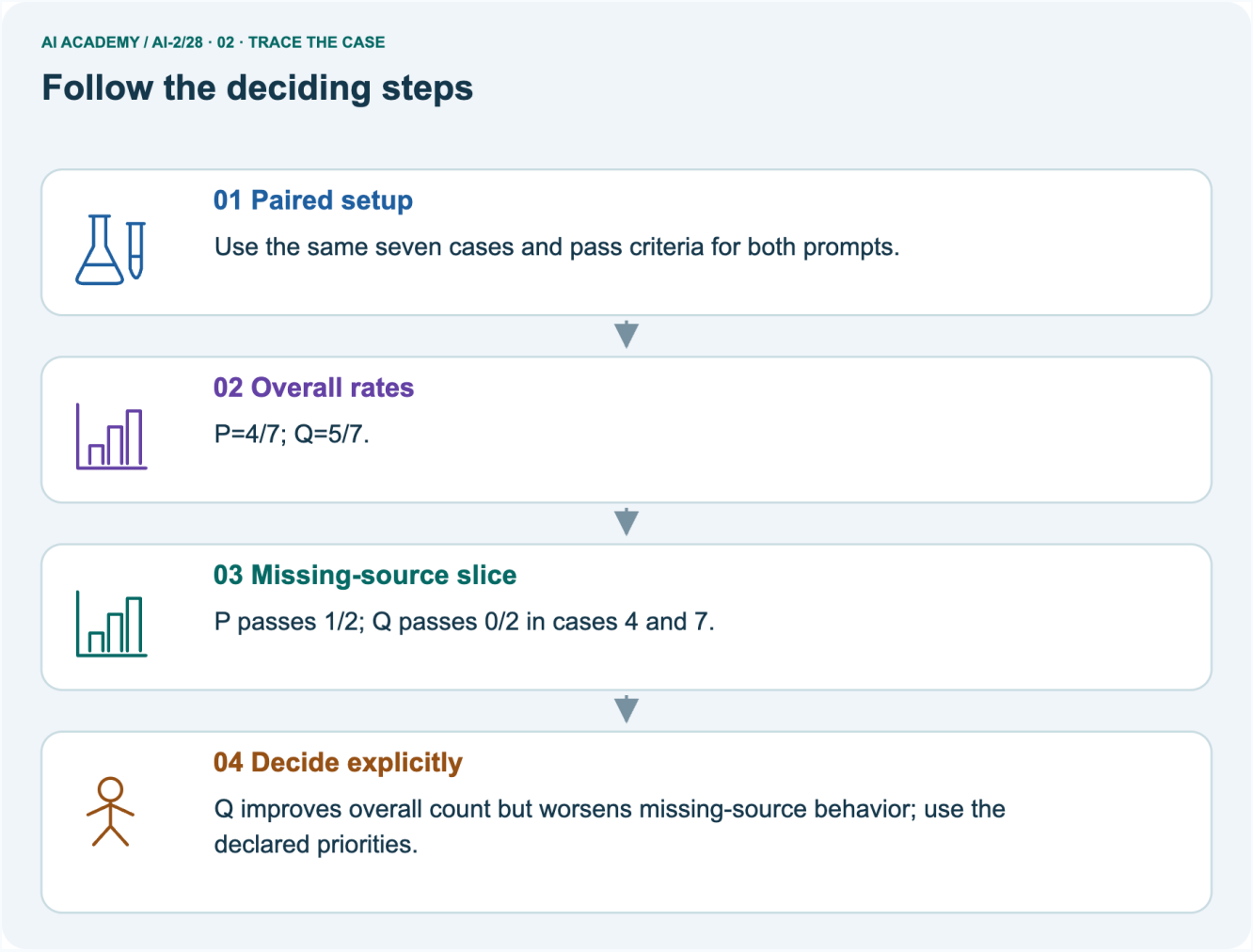
A fictional book-help project tests two **prompt** versions on seven shared development cases. Supplied facts: P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



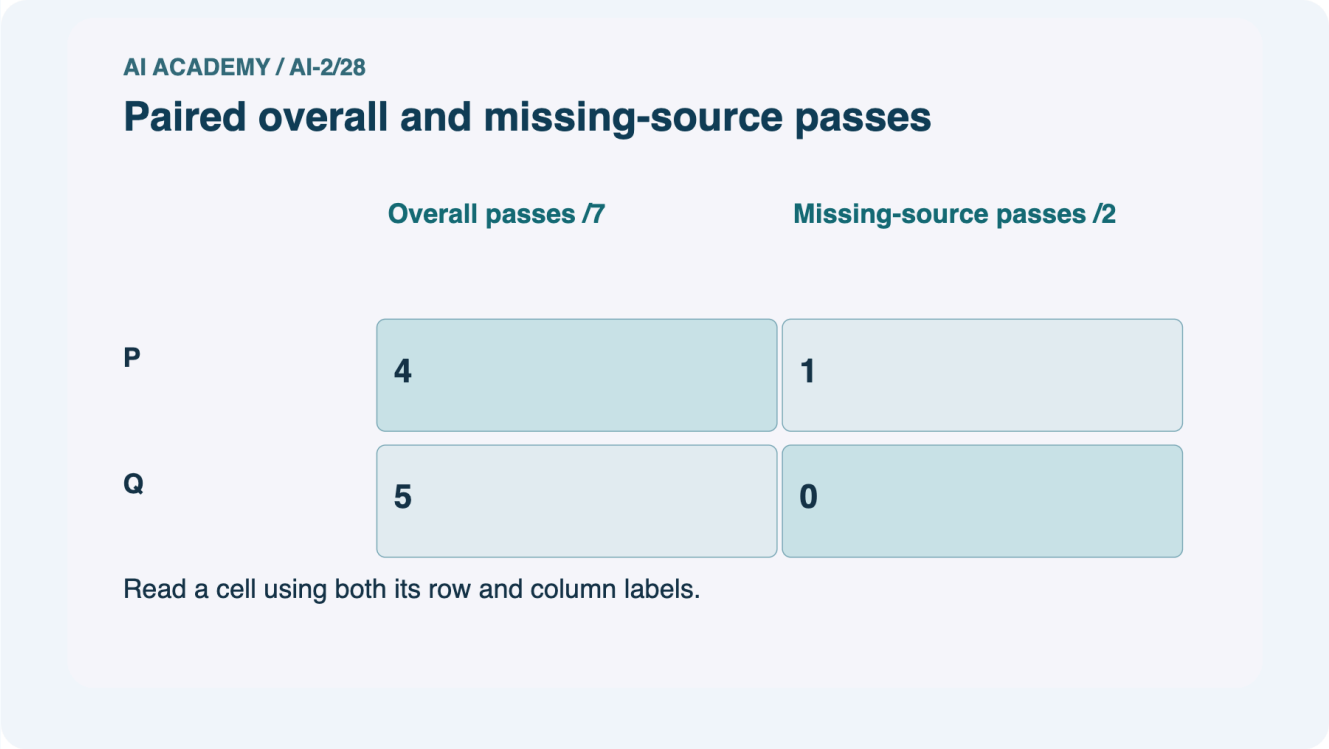
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired setup: Use the same seven cases and pass criteria for both **prompt**s. → Overall rates: P=4/7; Q=5/7. → Missing-source slice: P passes 1/2; Q passes 0/2 in cases 4 and 7. → Decide explicitly: Q improves overall count but worsens missing-source behavior; use the declared priorities.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Paired overall and missing-source passes



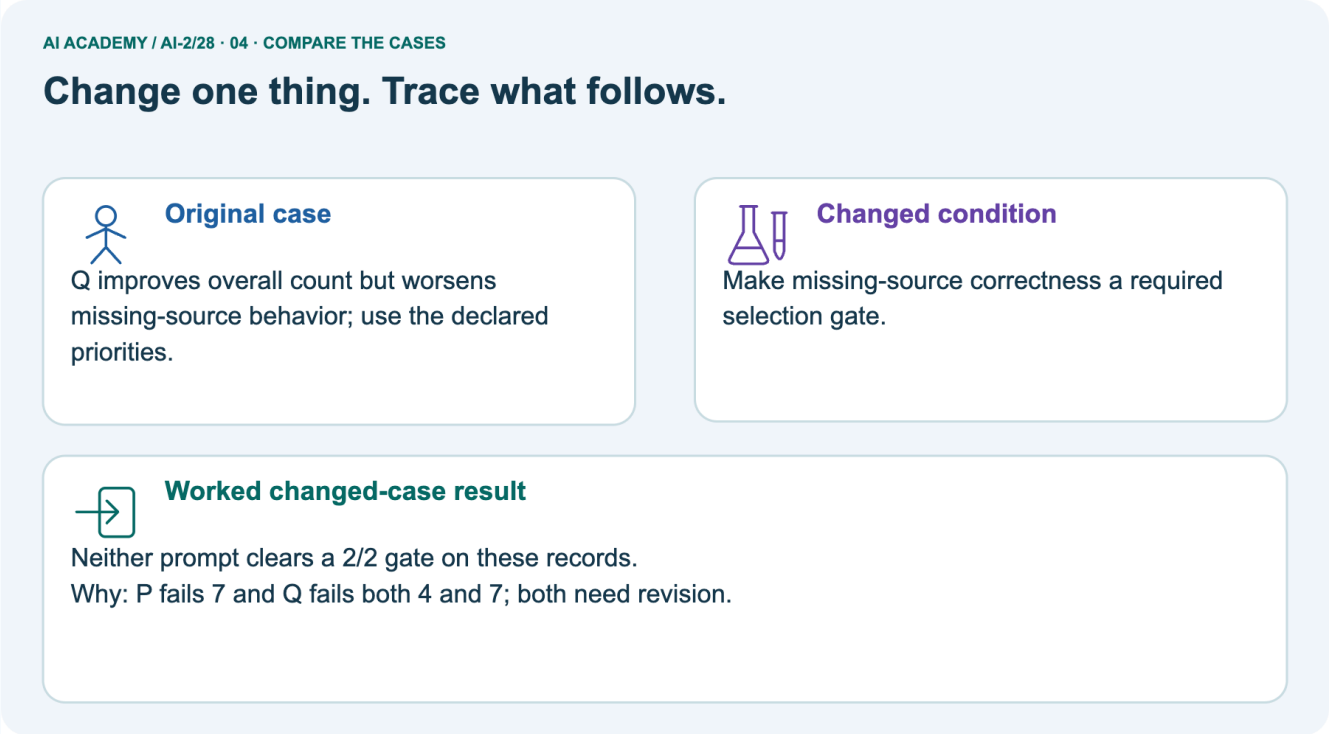
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired overall and missing-source passes. P: Overall passes /7 = 4, Missing-source passes /2 = 1. Q: Overall passes /7 = 5, Missing-source passes /2 = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

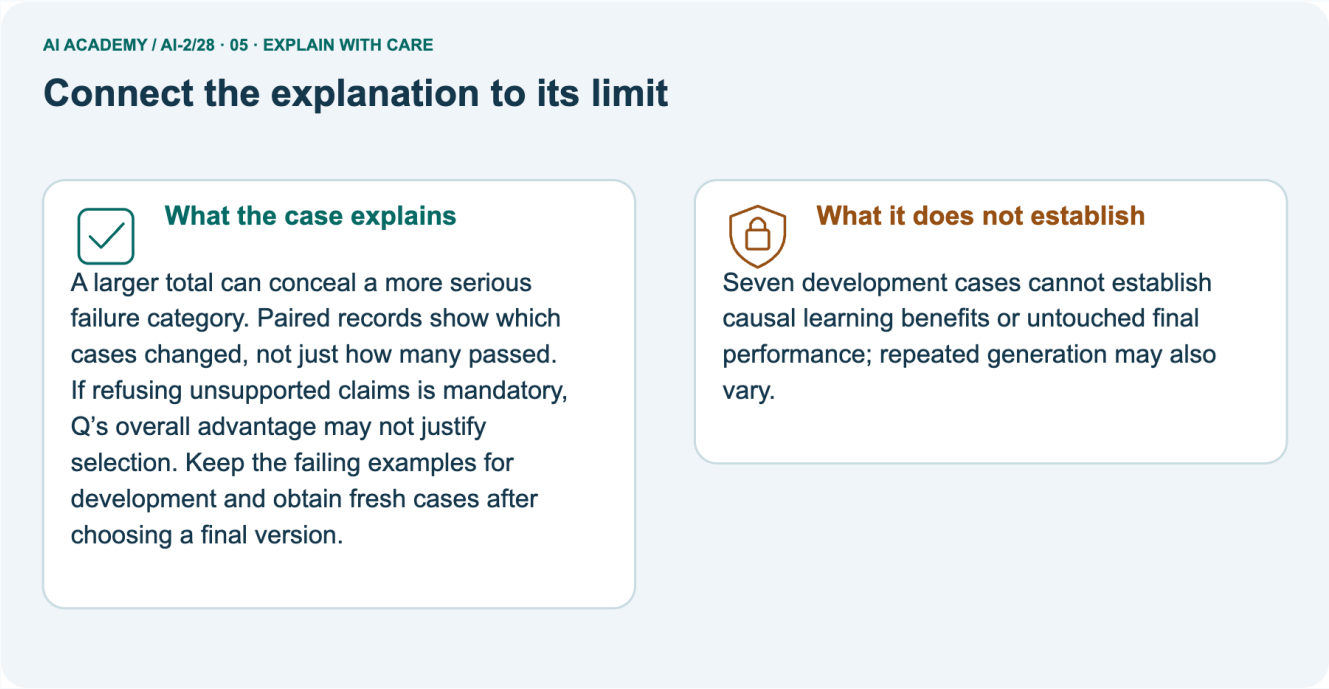
Original case: Q improves overall count but worsens missing-source behavior; use the declared priorities. Changed condition: Make missing-source correctness a required selection gate.

Expected result: Neither **prompt** clears a 2/2 gate on these records. Reason: P fails 7 and Q fails both 4 and 7; both need revision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

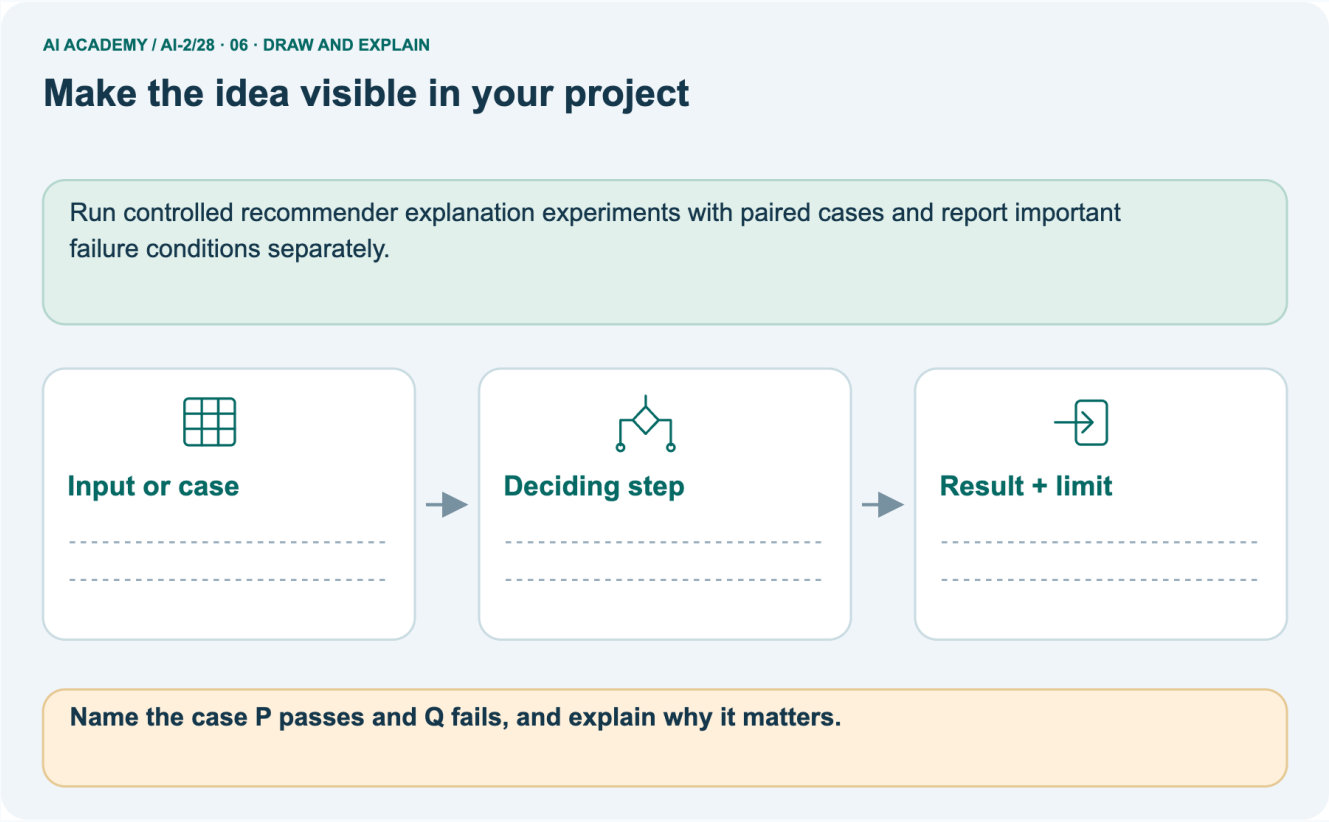
Keep the failing examples for development and obtain fresh cases after choosing a final version.

Limit: Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately. Draw your **input**, deciding step, result and limit.

Supported question: Name the case P passes and Q fails, and explain why it matters.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.